Macroeconomics I — MPE Insper

Mock Exam 4 (Simulado 4) — 2026, 3rd term

Instructions. Three questions, 10 points in total. Suggested time: 2 hours, closed book, non-programmable calculator allowed. Show every step: a number or a formula without the argument that produces it earns little. When an item says *explain*, *justify* or *show graphically*, the explanation and the graph are graded on their own lines: label the axes and every curve, mark the initial and the new point, and indicate the direction of every shift. Where an item lists several approaches or several choice variables, answer *each* of them. Box your final answers. Notation: Kurlat (2020) in Questions 1–2, Benigno (2015) in Question 3.

Question 1 (1.5 points)

A simplified economy has, at first, no government. Values are in R$ million for one year. A coffee farm in Minas Gerais (Firm F) grows beans worth 400 and sells all of them to a roastery (Firm R); F uses no inputs and pays 250 in wages. Firm R buys the 400 of beans and imports packaging worth 100 from abroad; it sells roasted coffee worth 700 to domestic households and worth 200 to foreign buyers, and pays 300 in wages. Each firm keeps whatever is left as profit.

1.a (0.5 point) Calculate the GDP of this economy by the product, the income and the expenditure approaches, explaining precisely the concept used in each. Explain why the imported packaging does not count in GDP, although Firm R used it to produce.

Solution:

1.b (0.5 point) Suppose instead that in the same year households buy only 600 of roasted coffee. Firm R produces exactly as before and keeps the remaining roasted coffee in its warehouse at the end of the year, valued at its selling price. Nothing else changes. Calculate, justifying, the GDP by the product, income and expenditure approaches, compare it with the value in 1.a and explain what the comparison indicates. Then explain what the sale of that stock to households in the following year adds to the following year's GDP, other things equal.

Solution:

1.c (0.5 point) Return to the situation of 1.a. The government now collects a lump-sum tax of 200 from households and uses it to: (i) pay 120 in wages to nurses in public clinics, which treat patients free of charge; (ii) buy 30 of imported medicines, handed out free at those clinics; (iii) pay 50 in cash transfers to low-income families (Bolsa Família). Households' purchases of roasted coffee stay at 700 and nothing else changes. Calculate, justifying, the GDP by the product, income and expenditure approaches, compare it with 1.a and explain what the comparison indicates. Explain why GDP does not rise by 200.

Solution:

Question 2 (5.5 points)

Consider an economy described by the Solow model, in which the aggregate variables satisfy:

$$Y_t = K_t^{\alpha}(A_t L_t)^{1-\alpha}, \tag{1}$$
$$K_{t+1} = (1-\delta)K_t + s\,Y_t, \tag{2}$$
$$L_{t+1} = (1+n)L_t, \tag{3}$$
$$A_{t+1} = (1+g)A_t, \tag{4}$$

with $\{\alpha, \delta, s\} \in (0,1)$ and $n, g \geq 0$.

2.a (1 point) Define the stationary variables, per efficiency unit of labour, with a tilde ($\tilde{k}_t$, $\tilde{y}_t$), and find an expression for $\tilde{k}_{t+1}/\tilde{k}_t$ and then the steady-state value of $\tilde{k}$, denoted $\tilde{k}_{SS}$. Show the steady state graphically. Along the steady state, at what rates do output per worker $y_t = Y_t/L_t$ and aggregate output Y_t grow? Justify, and explain why output per worker is not a stationary variable here.

Solution:

2.b (1 point) Show which saving rate maximises steady-state consumption per efficiency unit of labour (the golden-rule rate, s_{GR}). Explain why the same s_{GR} also maximises consumption per worker at every date along the steady state, and how a faster rate of technological progress g changes s_{GR} and $\tilde{k}_{GR}$. Give the economic intuition.

Solution:

2.c (1 point) Consider now the competitive equilibrium of an economy that lasts only two periods, $t = 1, 2$. There is an initial capital stock $K_1 > 0$, which belongs to the representative firm and depreciates fully once used in period 1. The representative firm maximises profits in perfectly competitive markets and has production function (1) in both periods, with $A_2 = (1 + g)A_1$. The representative consumer owns the firm and receives its profits Π_1 and Π_2. In each period the consumer has one unit of time, which it splits between work L_t and leisure $1 - L_t$, and earns real labour income $w_t A_t L_t$ (w_t is the real wage per efficiency unit of labour). Income not consumed in period 1 is saved as an asset a_1 that pays the real interest rate r; the only asset in this economy is the capital the firm uses in period 2, K_2, which also depreciates fully once used. Intertemporal utility is

$$U = \ln c_1 + \gamma \ln(1 - L_1) + \beta \left[\ln c_2 + \gamma \ln(1 - L_2)\right], \qquad \beta \in (0, 1),\ \gamma > 0.$$

Derive and explain the intertemporal budget constraint of this consumer.

Solution:

2.d (1 point) Set up the problem of the consumer of item 2.c and derive the first-order conditions for *all* of its choice variables, including the equation describing the intertemporal consumption decision and the consumption–leisure decision in each period. Then find the equation that gives the optimal c_1 as a function only of parameters, prices and levels of income and wealth, and interpret it economically.

Solution:

2.e (1 point) Calculate the saving rate of this representative consumer (the share of saving in the consumer's period-1 income), recalling that markets are perfectly competitive. Find the hours worked in each period. Then find an expression for $\hat{k}_2 \equiv K_2/A_2$ as a function only of $\hat{k}_1 \equiv K_1/A_1$ and of parameters. Treating this expression as a law of motion that applies period after period, draw it with $\hat{k}_1$ on the horizontal axis and $\hat{k}_2$ on the vertical axis, and show graphically the steady state of this economy. Is it stable? Justify.

Solution:

2.f (0.5 point) Discuss intuitively the relation between the model of items 2.c–2.e and the Solow model of items 2.a–2.b. In particular, compare the saving rate found in 2.e with s_{GR} and explain the difference.

Solution:

Question 3 (3 points)

Consider the simplified New Keynesian model of Benigno (2015), based on the aggregate demand and supply equations below (barred variables are long-run values, when all firms reset their prices; the others are short-run values):

$$y = \bar{y} + (g - \bar{g}) - \sigma\,[i - (\bar{p} - p)] - \sigma \ln \beta, \tag{AD}$$

$$p - p^e = \kappa\,(y - y_n), \tag{AS}$$

where p^e is the price set by the firms that cannot reset prices and y_n is short-run natural output,

$$y_n = \frac{1+\eta}{\sigma^{-1}+\eta}\,a + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}\,g - \frac{1}{\sigma^{-1}+\eta}\,\mu, \tag{YN}$$

μ being the mark-up; $\bar{y}$ is given by the same expression evaluated at the long-run values $\bar{a}$, $\bar{g}$, $\bar{\mu}$; and $y_e = \frac{1+\eta}{\sigma^{-1}+\eta}a + \frac{\sigma^{-1}}{\sigma^{-1}+\eta}g$ is the efficient level of output. All variables are log deviations from an efficient steady state. Initially $a = \bar{a} = g = \bar{g} = \mu = \bar{\mu} = 0$, $\bar{p} = p^e$, and the central bank sets $i = \rho \equiv -\ln\beta$, so that $y = y_n = y_e = \bar{y}$ and $p = p^e$.

3.a (1 point) News arrives that productivity will be permanently higher in the long run: $\bar{a}$ rises by $d\bar{a} > 0$, while current productivity a does not change. With i unchanged, represent graphically, explaining economically why, the initial equilibrium and the shift that defines the new equilibrium. Compute dy and dp. Indicate the gap $y - y_n$ and the gap $y - y_e$ and use both in your explanation. Is this boom good for welfare today? Justify.

Solution:

3.b (1 point) Based on the previous item, explain intuitively what a monetary authority committed to price stability should do after the news, by how much it must move its instrument, and what is the economic cost of doing so. Show graphically the paths of y and p in the short and in the long run, with and without that response. A commentator claims that "whatever the central bank does, it must give up either price stability or a closed output gap". Evaluate the claim.

Solution:

3.c (1 point) Suppose the monetary authority represents society's preferences by the quadratic loss function, with parameter $\phi \geq 0$,

$$L = \frac{1}{2}(y - y_e)^2 + \frac{\phi}{2\kappa}(p - p^e)^2 . \tag{LF}$$

In the same period as the news of item 3.a, the government temporarily raises a tax on firms' sales revenue (an IPI-like tax), which raises the current mark-up by $d\mu > 0$; the long-run mark-up $\bar{\mu}$ does not change. Derive the optimal monetary-policy rule (IT), find the optimal y, p and i, and show graphically the optimal policy. Discuss how your answer depends on ϕ, separating the part of the optimal response that answers the news from the part that answers the mark-up shock.

Solution:

End of exam.